

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Substitution: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{12} + \frac{1}{6}$ or $R = \frac{12 \times 6}{12 + 6}$ (1) $R = 4 \text{ } [\Omega]$ (1) $R = R_1 + R_2 = 4$ (ecf only from use of parallel equation) + 16 = 20 $[\Omega]$ (1) Answer of 16.25 $[\Omega]$ award 2 marks	1	1 1		3	3	3
	(b)			$P = I^2 R = 1.5^2 \times 20$ (ecf) = 45 $[\text{W}]$ (1) Substitution: $t = \frac{75.6 \times 10^3}{45}$ (ecf) (1) = 1 680 $[\text{s}]$ (1) $\frac{1680(\text{ecf})}{60} = 28$ $[\text{minutes}]$ (1) Answer of 1.68×10^n where n is not equal to 3 award 2 marks Answer of 2.8×10^n where n is not equal to 1 award 3 marks	1	1 1 1		4	4	4
				Question 5 total	2	5	0	7	7	7